



Techniques of Circuit Analysis

1. Node Voltage Method (Nodal Analysis)

Nodal analysis is a systematic method of circuit analysis. In this method node voltages are taken as the circuit variables.

A) For a circuit with n nodes and **without any voltage sources**, take these steps:

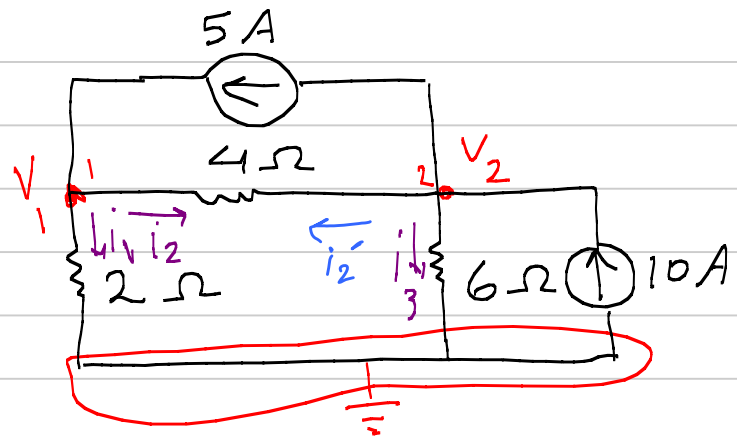
1. Select one node as the reference point, assign node voltages for the remaining nodes. (These are the voltages of the nodes with respect to the reference node).

2. Apply KCL at each node. Use Ohm's law to express all currents in terms of the node voltages.

3. Solve the KCL equations to find the node voltages.

(#of unknowns = $n-1$)

Ex. Find the node voltages.



$$\text{at } \textcircled{1}: i_1 + i_2 - 5 = 0$$

$$\frac{V_1}{2} + \frac{V_1 - V_2}{4} - 5 = 0 \quad \textcircled{*}$$

$$\text{at } \textcircled{2}: -i_2 + i_3 + 5 - 10 = 0$$

$$-\frac{V_1 - V_2}{4} + \frac{V_2}{6} - 5 = 0 \quad \textcircled{**}$$

$$\text{or: } i_2 + i_3 + 5 - 10 = 0$$

$$\frac{V_2 - V_1}{4} + \frac{V_2}{6} - 5 = 0$$

This is identical to ~~the first equation~~.

Therefore, we can assume all resistor currents are leaving the node if the currents are expressed in terms of voltages.

$$\textcircled{*} : 3V_1 - V_2 = 20$$

$$\textcircled{**} : -3V_1 + 5V_2 = 60$$

$$\begin{bmatrix} 3 & -1 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 60 \end{bmatrix}$$

(in matrix form)

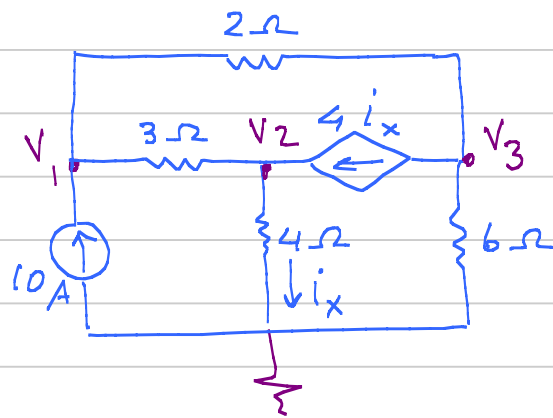
Cramer's method:

$$\Delta = \begin{vmatrix} 3 & -1 \\ -3 & 5 \end{vmatrix} = 12$$

$$\Delta_1 = \begin{vmatrix} 20 & -1 \\ 60 & 5 \end{vmatrix} = 160 \Rightarrow V_1 = \frac{\Delta_1}{\Delta} = \underline{13.33 \text{ V}}$$

$$\Delta_2 = \begin{vmatrix} 3 & 20 \\ -3 & 60 \end{vmatrix} = 180 + 60 = 240 \rightarrow V_2 = \frac{\Delta_2}{\Delta} = \underline{20 \text{ V}}$$

Ex. Find the node voltages.



$$-10 + \frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{2} = 0 \quad (1)$$

$$\frac{V_2 - V_1}{3} + \frac{V_2}{4} - 4 \left(\frac{V_2}{4} \right) = 0 \quad (2)$$

$$\frac{V_3}{6} + \frac{V_3 - V_1}{2} + 4 \left(\frac{V_2}{4} \right) = 0 \quad (3)$$

$$V_1 = 80V$$

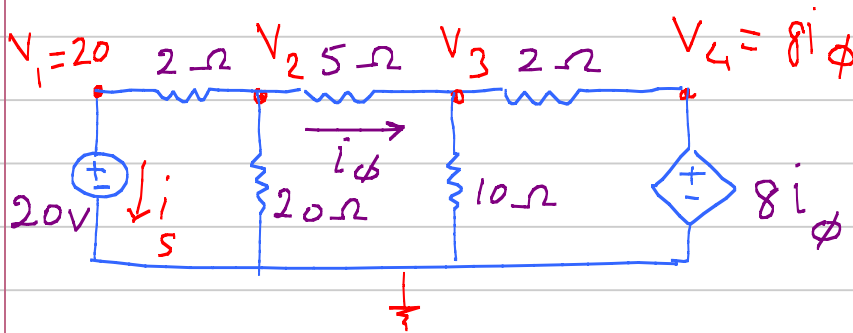
$$V_2 = -64V$$

$$V_3 = 156V$$

B) when a **voltage source exists in the circuit**, there are two possibilities:

case1: If the **voltage source is connected between a node and the reference node**, the voltage of that node is known and equals the voltage of the source.

Ex. Find the power associated with the 20v source using nodal analysis.



$$\text{KCL at node 2: } \frac{V_2 - 20}{2} + \frac{V_2}{20} + \frac{V_2 - V_3}{5} = 0$$

$$\text{KCL at node 3: } \frac{V_3 - V_2}{5} + \frac{V_3}{10} + \frac{V_3 - 8\left(\frac{V_2 - V_3}{5}\right)}{2} = 0$$

$$V_2 = \underline{16V}, \quad V_3 = \underline{10V}$$

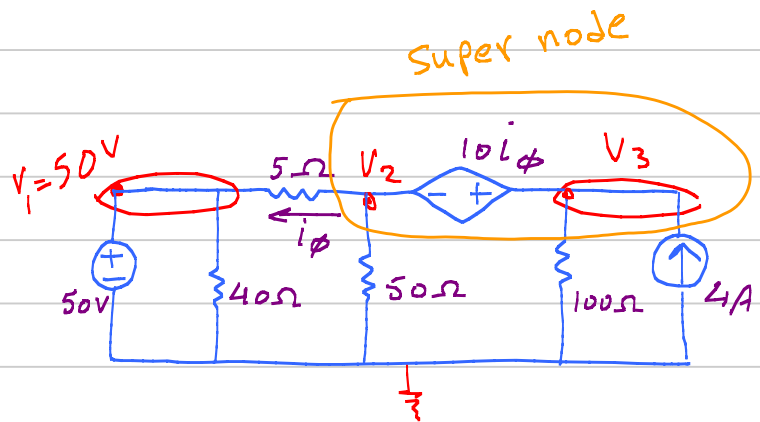
$$i_s = \frac{V_2 - 20}{2} = -2A$$

$$P_{20V} = V i_s = (20)(-2) = \underline{-40W}$$

case 2: If the voltage source is connected between two non-reference nodes, combine the two nodes to form a super node. Apply KCL at the super node. The difference between the two voltages equals the voltage of the source (constraint equation).

A supernode should enclose the voltage source and any other elements in parallel with it.

Ex. Find i_ϕ .

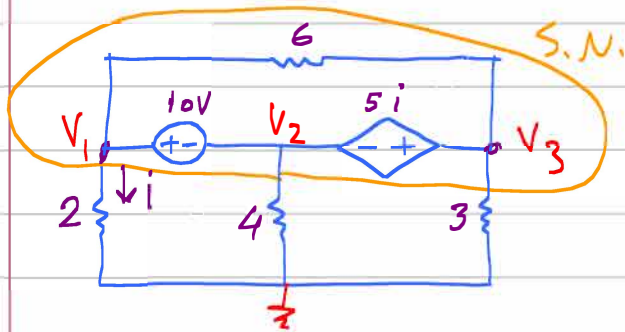


$$\text{KCL at SN : } \frac{V_2 - 50}{5} + \frac{V_2}{50} + \frac{V_3}{100} - 4 = 0 \quad (1)$$

$$\text{constraint equation: } V_3 - V_2 = 10 \left(\frac{V_2 - 50}{5} \right) \quad (2)$$

$$V_2 = 60V, V_3 = 80V \Rightarrow i_\phi = \frac{V_2 - 50}{5} = 2A$$

Ex. Find the node voltages. Try different reference points.



KCL at S.N. :

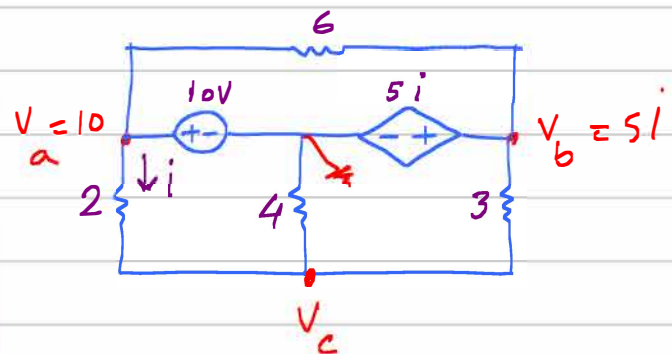
$$\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} = 0 \quad (1)$$

constraint equations:

$$V_2 - V_1 = 10 \quad (2)$$

$$V_3 - V_2 = 5\left(\frac{V_1}{2}\right) \quad (3)$$

$$\Rightarrow \begin{cases} V_1 = 3.04 \text{ V} \\ V_2 = -6.96 \text{ V} \\ V_3 = 0.65 \text{ V} \end{cases}$$



KCL at node 3 :

$$\frac{V_c - 10}{2} + \frac{V_c}{4} + \frac{V_c - 5\left(\frac{10 - V_c}{2}\right)}{3} = 0$$

$$V_c = 6.96$$

$$V_b = 5i = 5\left(\frac{10 - 6.96}{2}\right) = 7.61$$

$$\begin{array}{l} V_1 \longleftrightarrow V_a - V_c \\ 3.04 \longleftrightarrow 10 - 6.96 \quad \checkmark \end{array}$$

$$\begin{array}{l} V_2 \longleftrightarrow -V_c \\ -6.96 \longleftrightarrow -6.96 \quad \checkmark \end{array}$$

$$\begin{array}{l} V_3 \longleftrightarrow V_b - V_c \\ 0.65 \longleftrightarrow 7.61 - 6.96 \quad \checkmark \end{array}$$